Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

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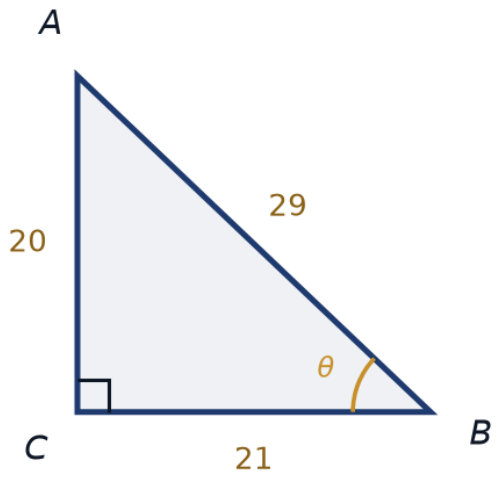
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a circle of radius $10\ \text{cm}$, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

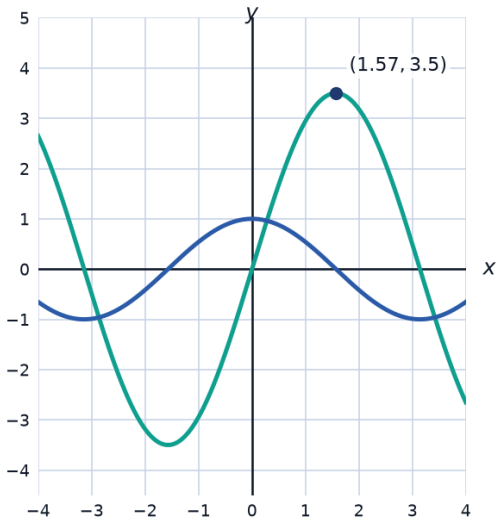
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A small particle of weight $24\ \text{N}$ is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x \to 0} \frac{1 - \cos(3x)}{x^2}$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

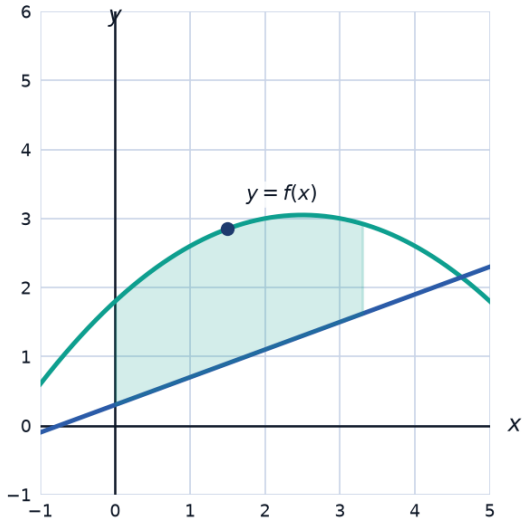
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A surveyor measures distances from an observation point C to two landmarks A and B: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance AB between the landmarks to the nearest tenth of a meter, and compute the area of ΔABC to the nearest square meter.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

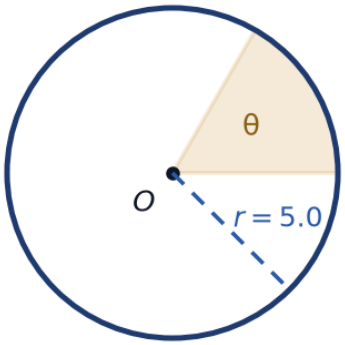
[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x \to 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3}$.

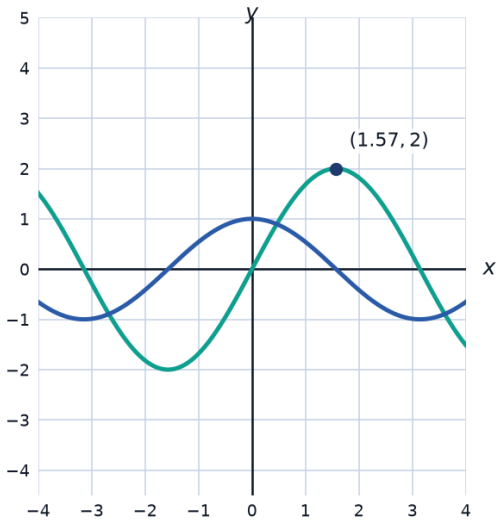
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem**

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.



17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 The Poisson Distribution

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Further Integration Techniques

A stone is thrown vertically upwards with speed 15 m/s from the top of a cliff 40 m above sea level. Taking $g = 9.8$ m/s^2 and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the trigonometric expression $\frac{\sec x - \cos x}{\tan x}$.

a. $\sin x$

b. $\cos x$

c. $\cot x$

d. $\csc x$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the exact value of $\csc\left(\frac{5\pi}{6}\right)$.

a. 2

b. -2

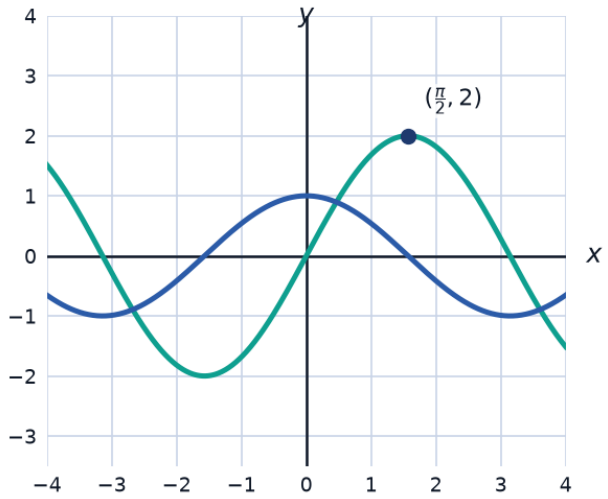
c. $\frac{2}{\sqrt{3}}$

d. $-\frac{2}{\sqrt{3}}$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which identity relates $\cot\theta$ and $\csc\theta$ for all defined angles?



a. $1 + \cot^2\theta = \csc^2\theta$

b. $1 + \csc^2\theta = \cot^2\theta$

c. $\cot^2\theta - \csc^2\theta = 1$

d. $\csc\theta = 1 + \cot\theta$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use compound angle identities to find the exact value of $\cos(75^\circ) = \cos(45^\circ + 30^\circ)$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d. $\frac{\sqrt{2}-\sqrt{6}}{4}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\sin\theta = \frac{3}{5}$ and θ is acute, what is $\cos(2\theta)$?

a. $\frac{7}{25}$

b. $\frac{24}{25}$

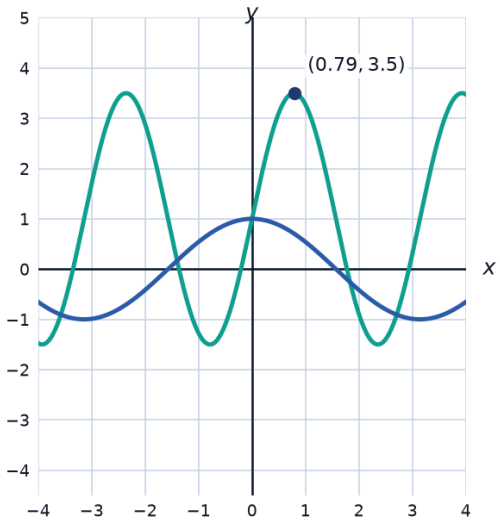
c. $-\frac{7}{25}$

d. $\frac{16}{25}$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Under the same conditions ($\sin\theta = \frac{3}{5}$ acute), what is $\sin(2\theta)$?



a. $\frac{24}{25}$

b. $\frac{7}{25}$

c. $\frac{12}{25}$

d. $\frac{6}{5}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express $\cos^2 x$ in terms of a first-power cosine function.

a. $\frac{1+\cos(2x)}{2}$

b. $\frac{1-\cos(2x)}{2}$

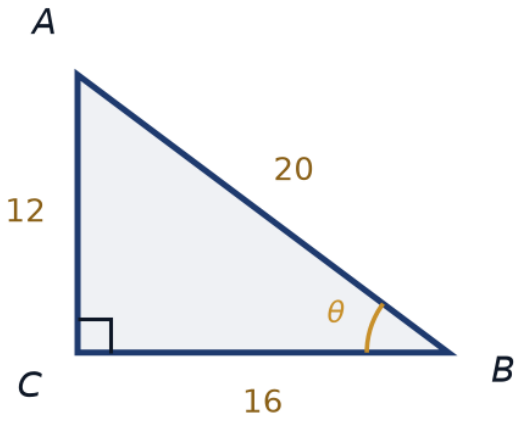
c. $\frac{1+\sin(2x)}{2}$

d. $1-\sin^2 x$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express $3\sin x + 4\cos x$ in the form $R\sin(x + \alpha)$ with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.



a. $5\sin(x + 0.927)$

b. $5\sin(x + 0.643)$

c. $7\sin(x + 0.927)$

d. $25\sin(x + 0.927)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the maximum value of the function $f(x) = 5\cos x - 12\sin x + 3$?

a. 16

b. 13

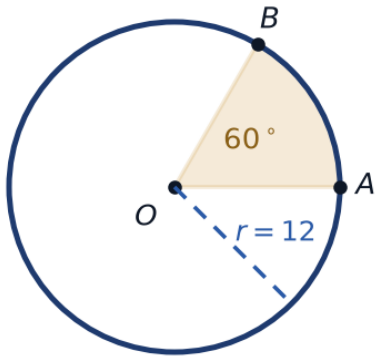
c. 10

d. 20

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

c. $\tan\theta$

d. 1

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the reference angle for 220°.

a. 60°

b. 40°

c. 50°

d. 140°

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The table below shows the results of a survey on preferred music genres among a group of students. What is the relative frequency of students who prefer rock music?

a. 0.20

b. 0.25

c. 0.30

d. 0.40

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

A small study was conducted on the number of hours students spent studying per week. The results are summarized in the frequency table below: | Hours Studied (h) | Frequency (f) | | :---------- | :---------- | | $0 \le h < 5$ | 12 | | $5 \le h < 10$ | 25 | | $10 \le h < 15$ | 18 | | $15 \le h < 20$ | 15 | What is the total number of students surveyed?

a. 60

b. 70

c. 65

d. 75

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 The Poisson Distribution

Evaluate $\int_0^2 xe^{2x}\,dx$ (in terms of e).

a. $\frac{3}{4}e^4 - \frac{1}{4}$

b. $\frac{3}{4}e^4$

c. $\frac{3}{4}e^4 + \frac{1}{4}$

d. $\frac{1}{4}e^4 + \frac{3}{4}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Further Integration Techniques

A 10-ft ladder leans against a wall; the base is 6 ft from the wall and slides away at $\frac{dx}{dt} = 1$ ft/s. Find $\frac{dy}{dt}$, the rate the top slides down (height $y = 8$ ft).

a. -1 ft/s

b. $-\frac{3}{4}$ ft/s

c. $-\frac{1}{4}$ ft/s

d. $-\frac{1}{2}$ ft/s

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Differential Equations

A graph has edges $1-2, 2-3, 3-4, 4-1$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Euler circuit exists

b. Euler trail exists, but not an Euler circuit

c. Cannot be determined from the given information

d. Neither an Euler trail nor circuit exists

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A weighted graph has edges (with weights) $1{-}2(1), 2{-}3(6), 3{-}4(5), 4{-}5(7), 1{-}5(8), 2{-}5(2), 1{-}3(9)$. Find the total weight of the minimum spanning tree.

a. 18

b. 11

c. 10

d. 14

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

a. 0.383

b. 0.533

c. 0.333

d. 0.433

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Matrices (AI HL)

Given $P(A) = \frac{2}{5}$, $P(B|A) = \frac{1}{2}$, and $P(B|A') = \frac{3}{10}$, find $P(A|B)$ to 3 decimal places.

a. 0.426

b. 0.576

c. 0.526

d. 0.476

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

If $\frac{dy}{dt} = \frac{1}{4}y$ and $y(0) = 9$, find $y(4)$ in terms of e.

a. $9 + e$

b. $9e^{-1}$

c. $9e$

d. $9\frac{1}{4}e$

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